

Srikanth Guduri

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PROFESSIONAL SUMMARY

Senior Software Engineer with 5+ years building scalable full-stack, distributed, and AI-powered applications for government, retail, and fintech platforms. Expert in Java, Spring Boot, microservices, Kafka, and event-driven architectures; skilled with React, TypeScript, and Node.js for frontend integration. Deep expertise in Apache Spark for large-scale data pipelines, SQL optimization across Oracle and PostgreSQL, and cloud-native deployment on AWS (Lambda, ECS, EKS, S3) with Terraform. Proficient in Docker, Kubernetes, and CI/CD automation. Experienced applying Generative AI, prompt engineering, RAG, vector databases, semantic search, and LLM evaluation to build intelligent automation and AI-assisted development solutions. Proven track record architecting resilient systems, optimizing performance at scale, mentoring engineers, and establishing secure coding and responsible AI practices.

SKILLS

Programming Languages: Java 8/11/17, Python, JavaScript, TypeScript, SQL

Frontend: React.js, Node.js, TypeScript, HTML5, CSS3, Responsive UI Development, Frontend Optimization

Backend: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, RESTful APIs, Microservices Architecture, API Design, Secure API Development

AI/ML & Generative AI: Generative AI: Prompt Engineering, RAG, AI Agents, Semantic Search, Vector Databases, LLM Evaluation & Output Validation, Azure AI Foundry, AI-Assisted Coding, AI-Assisted Development, Knowledge Retrieval Systems, Intelligent Automation, Responsible AI, Secure AI Usage

Cloud, DevOps & Big Data: AWS (S3, EC2, Lambda, ECS, EKS, CloudWatch), Databricks, Apache Spark, Docker, Kubernetes, Terraform, Jenkins, CI/CD Pipelines, Containerization, Deployment Automation

Databases, Messaging & Caching: PostgreSQL, Oracle, SQL Server, MySQL, Redis, Apache Kafka, Event Streaming, Event-Driven Architecture, Distributed Caching, SQL Optimization, Query Optimization

Distributed Systems & Performance Engineering: Distributed Systems, Event-Driven Architecture, Asynchronous Processing, Parallel Compute Strategies, Scalability Engineering, Fault Tolerance, Retry Mechanisms, Resiliency Patterns, High-Throughput Systems, Low-Latency Systems, Telemetry Processing, Performance Optimization

Software Engineering & System Design: System Design, HLD/LLD, OOP, Data Structures & Algorithms, Multithreading, Concurrency, Production Debugging, Root Cause Analysis, Technical Ownership, Feature Ownership, End-to-End Ownership, Micro Frontends

Testing & Development: JUnit 5, Mockito, Unit Testing, Integration Testing, Test Automation, Code Reviews, Agile, Scrum, SDLC, Sprint Planning

Leadership: Cross-Functional Collaboration, Stakeholder Management, Technical Mentorship, Engineering Excellence, Problem Solving

WORK EXPERIENCE

NYC Office of Technology and Innovation | Senior Software Engineer

Aug 2023 - Present

- Built Kafka event pipelines decoupling application-submitted and status-changed events across NYC Business Portal microservices, eliminating tight coupling and reducing cascade failures.
- Optimized 40+ slow PostgreSQL queries for reporting workflows, restructuring catalog scans to cut unnecessary I/O; improved query latency by 70% and unblocked 100+ daily citizen requests.
- Developed Java Spring Boot REST APIs for licensing, permits, and inspections modules on NYC Business Portal, handling 50K+ concurrent citizen workflows at scale.
- Consolidated multi-source agency data into single cleaned dataset using Databricks batch pipelines, enabling downstream reporting and reducing manual ETL overhead by 80%.
- Deployed Spring Boot services to AWS Lambda and EC2 based on workload patterns using Terraform; reduced cold-start latency by 40% through infrastructure optimization.
- Built and shipped React/TypeScript frontend components end-to-end, integrating with backend REST APIs to deliver complete citizen-facing features within sprint cycles.
- Decomposed monolithic portal frontend into independently deployable Micro Frontends, enabling 5+ teams to release features in parallel without coordination overhead.
- Accelerated Java refactoring and unit-test creation using AI-assisted tools; established peer-review and SonarQube validation gates, reducing defect escape rate by 35%.
- Streamlined developer onboarding by documenting API contracts, architecture patterns, and deployment procedures; reduced time-to-first-commit for new engineers from 4 weeks to 1 week.
- Built AI-assisted code review and debugging workflows using LLMs with prompt engineering and RAG; validated AI-generated suggestions against test suites and deployed fixes with 98% accuracy.

Michelin | Software Development Engineer

Feb 2021 - Jul 2022

- Designed Spark jobs consolidating 5+ retail warehouse data sources into unified analytics dataset, reducing batch runtime from 3 hours to 45 minutes and enabling real-time catalog insights for merchandisers.
- Built Kafka producer/consumer pipelines processing 100K+ real-time inventory updates daily from warehouse systems; decoupled producers from slower consumers, eliminating cascading delays.
- Developed Java Spring Boot microservices for product catalog, inventory, and personalization domains, handling 200K+ daily transactions across Michelin's retail platform.
- Optimized slow Oracle product search query scanning 40% of catalog unnecessarily; restructured joins and indexes to cut search latency by 55%, improving storefront response time.
- Architected Node.js WebSocket service layer for real-time inventory status, pushing live stock updates to 50K+ concurrent storefronts and eliminating polling overhead.
- Contributed to Micro Frontend implementation enabling 4 teams to own and deploy independent storefront sections; reduced deployment coordination meetings by 70%.
- Deployed containerized services to AWS ECS and EKS, evaluating each based on workload patterns and operational complexity; reduced infrastructure costs by 25% through right-sizing.
- Instrumented services across Splunk, Dynatrace, Grafana, and Datadog using OpenTelemetry; achieved 99.5% observability coverage and reduced MTTR (mean time to resolution) by 40%.

Paytm | Software Engineer

Dec 2019 - Jan 2021

- Developed and integrated Spring Boot REST APIs for account management and transaction processing, handling 1M+ daily transactions with secure authentication and authorization controls to protect sensitive financial data.
- Optimized Oracle and SQL Server queries for reporting; restructured slow queries to cut execution time by 45%, improving end-of-day batch performance and unblocking reporting workflows.
- Implemented Spring Security authentication and role-based authorization across 8+ microservices, protecting sensitive financial transaction data and enforcing access control policies.
- Developed and deployed backend services using Maven for builds, Git for version control, and Jenkins CI/CD pipelines; reduced manual deployment time from 2 hours to 15 minutes.
- Created comprehensive JUnit unit tests validating application functionality and reliability; established testing standards that improved code quality and reduced production defects.

CERTIFICATIONS & ACHIEVEMENTS

- AWS Certified Developer – Associate (DVA-C02) - AWS
- Microsoft Certified: Azure AI Apps and Agents Developer Associate (AI-103) - Microsoft

EDUCATION

University of Central Missouri

Master of Science, *Computer Science*

Dec 2023

KL University

Bachelor of Technology, *Computer Science*

May 2020